

Companion XL: CMB Polarisation in the Relaxation-Driven Cyclic Cosmology

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Abstract

The polarisation of the cosmic microwave background (CMB) encodes the temporal structure of the gravitational field at the epoch of recombination and during the subsequent evolution of the Universe. In the standard cosmological model, E-mode polarisation arises from scalar perturbations, while B-modes are generated by tensor perturbations or by lensing of E-modes. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the quadrupole anisotropy that sources polarisation, leading to characteristic signatures in both E- and B-modes. This Companion develops a continuous description of CMB polarisation in RDCC, deriving how the relaxation scale influences the generation of E-modes, the leakage into B-modes, and the late-time evolution of polarisation through the integrated Sachs–Wolfe effect. The resulting framework provides a new observational window into the temporal architecture of the RDCC gravitational field.

1 Introduction

CMB polarisation arises from Thomson scattering of photons off free electrons in the presence of a quadrupole temperature anisotropy. The structure of this quadrupole is determined by the evolution of the gravitational potential and the velocity field of the photon–baryon fluid. In the standard cosmological model, the potential evolves smoothly, and the resulting polarisation pattern is dominated by E-modes, with B-modes generated only by tensor perturbations or by lensing.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the quadrupole anisotropy at recombination and during the subsequent evolution of the Universe. As a result, the polarisation pattern acquires a distinctive RDCC signature that reflects the temporal structure of the gravitational field.

2 The RDCC Gravitational Potential and the Quadrupole Source

The generation of CMB polarisation is governed by the quadrupole moment of the photon distribution. This quadrupole is sourced by the time evolution of the gravitational potential. In RDCC, the potential takes the form

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (1)$$

The quadrupole source term becomes

$$\Theta_{2,\text{RDCC}}(k, a) = \Theta_2(k, a) - \chi_2 \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \frac{\partial \Phi}{\partial a}. \quad (2)$$

Small-scale modes contribute less to the quadrupole, while large-scale modes retain their coherence. This imbalance reshapes the polarisation pattern at recombination.

3 E-Mode Polarisation in RDCC

E-mode polarisation arises from the curl-free component of the polarisation field. In RDCC, the suppression of small-scale potential evolution reduces the amplitude of high- ℓ E-modes. The E-mode power spectrum becomes

$$C_{\ell}^{EE, \text{RDCC}} = C_{\ell}^{EE} \left[1 - \chi_E \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (3)$$

where ℓ_{rel} corresponds to the angular scale of the relaxation scale k_{rel} .

This modification produces a smoother E-mode spectrum with reduced small-scale power and enhanced large-scale coherence.

4 B-Mode Polarisation and RDCC-Induced Leakage

B-mode polarisation is generated by tensor perturbations or by lensing of E-modes. In RDCC, the altered evolution of the gravitational potential introduces an additional source of B-modes. The leakage from E- to B-modes becomes

$$C_{\ell}^{EB, \text{RDCC}} \propto \left(\frac{\partial \Phi_{\text{RDCC}}}{\partial a} \right) C_{\ell}^{EE}. \quad (4)$$

Because the relaxation mechanism modifies the time derivative of the potential, this leakage acquires a scale dependence that reflects the relaxation scale. The resulting B-mode spectrum is not dominated by primordial tensor modes but by the temporal structure of the RDCC gravitational field.

This provides a new observational signature that distinguishes RDCC from the standard cosmological model.

5 Late-Time Evolution and the Integrated Sachs–Wolfe Contribution

The integrated Sachs–Wolfe (ISW) effect contributes to CMB polarisation through the late-time evolution of the gravitational potential. In RDCC, the potential evolves differently at late times, leading to a modified ISW contribution to both E- and B-modes.

The ISW-induced polarisation becomes

$$\Delta P_{\text{ISW}, \text{RDCC}} \propto \int da \frac{\partial \Phi_{\text{RDCC}}}{\partial a} \Theta_1(a), \quad (5)$$

where Θ_1 is the dipole moment of the photon distribution.

Because the relaxation mechanism suppresses the decay of small-scale modes, the ISW contribution becomes smoother and more coherent, enhancing large-scale polarisation.

6 Conclusion

CMB polarisation in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the quadrupole anisotropy at recombination, the generation of E-modes, the leakage into B-modes, and the late-time ISW contribution. These effects produce distinctive signatures in the polarisation power spectra that

provide a direct observational probe of the relaxation scale and the temporal structure of the gravitational field. The present Companion establishes the theoretical foundation for future studies of CMB secondary anisotropies, lensing-induced polarisation, and the interplay between scalar and tensor modes in RDCC.

References

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